

STATUS OF THE MØLLER POLARIMETER

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Hall A Møller Polarimeter

In operation since 1997

Original energy range: $0.8 \div 4.0$ GeV

Length along the beam line : 7m

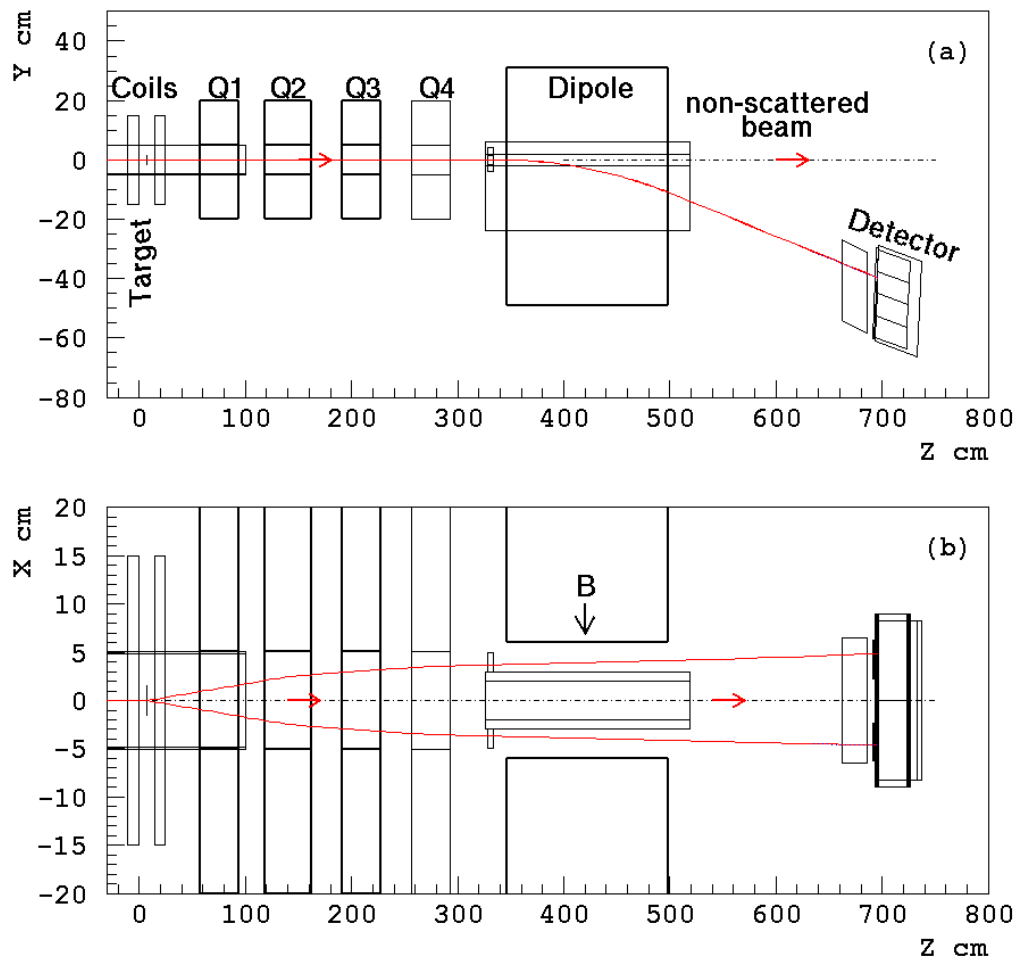
Accuracy in 1997: 3.5%

After 11 GeV upgrade

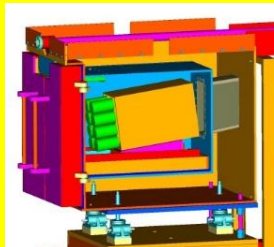
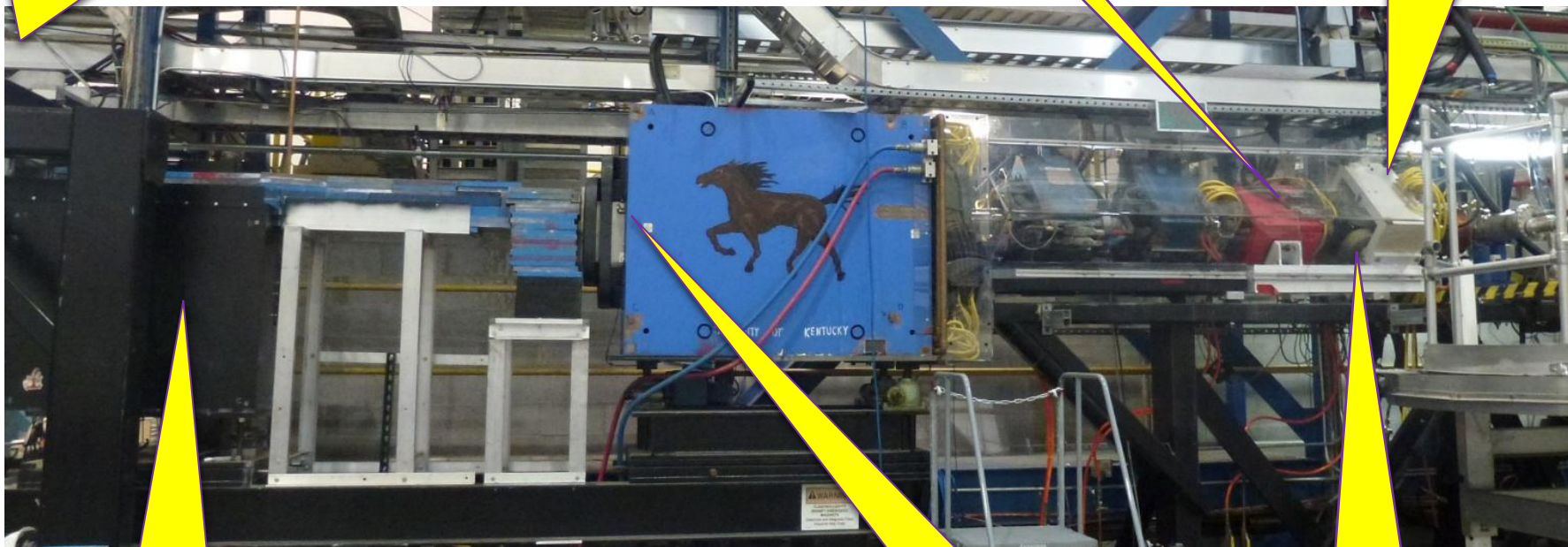
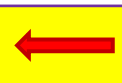
$0.8 \div 11.0$ GeV

7m

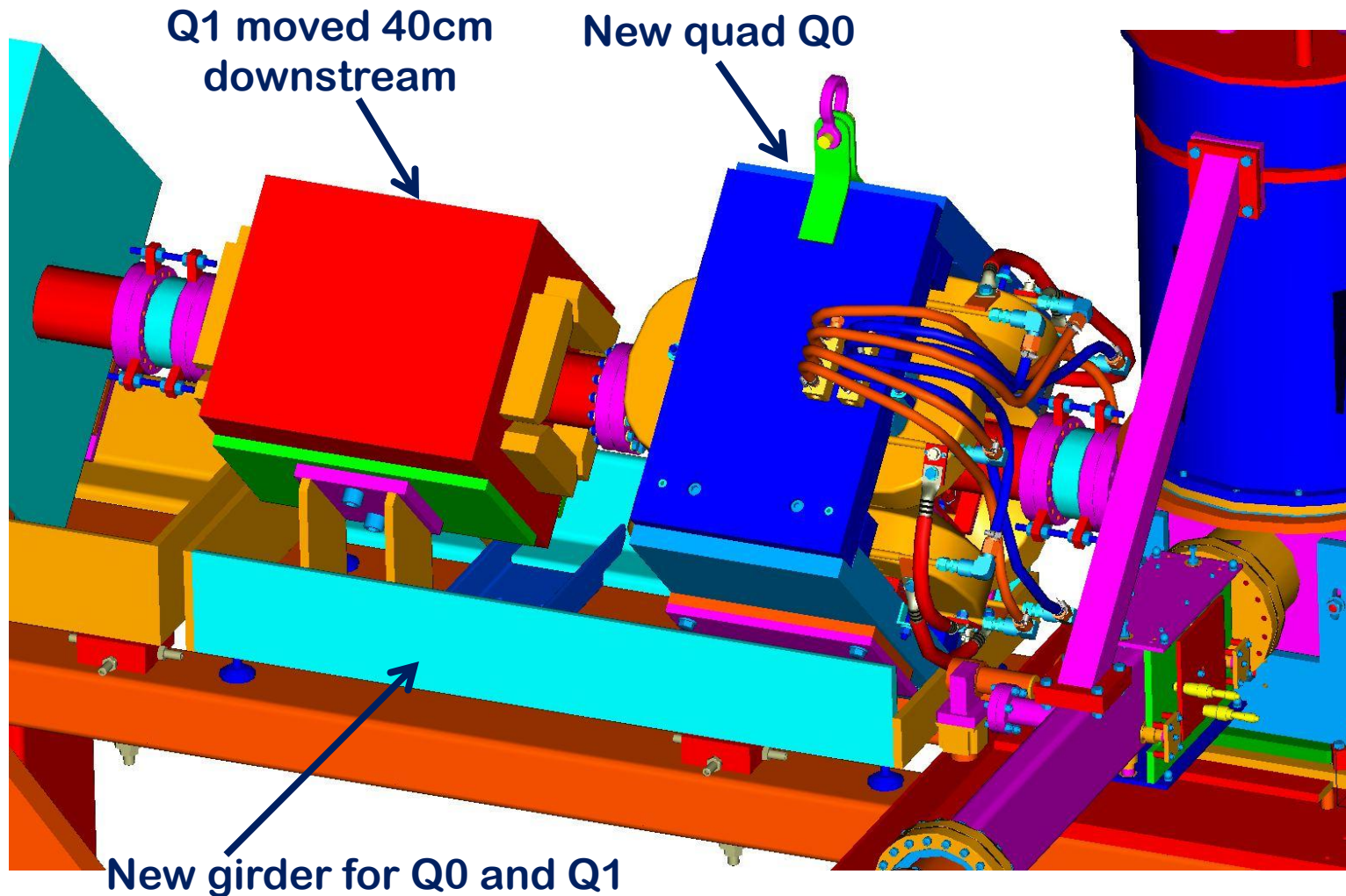
2013: $0.9\% \div 1.7\%$



11 GeV Upgrade

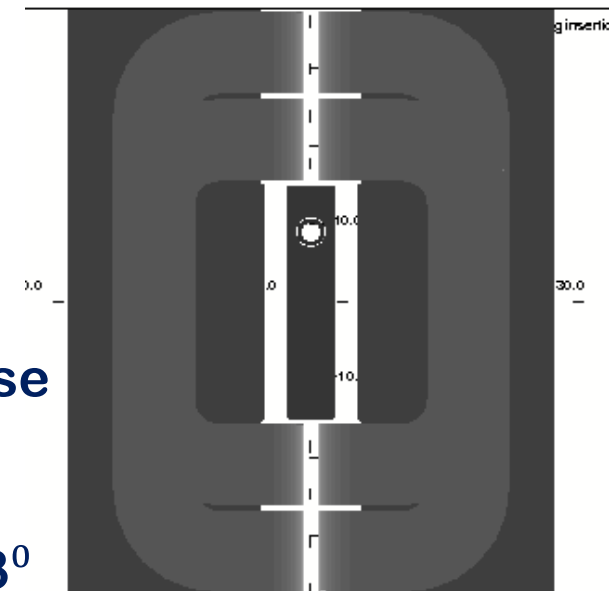
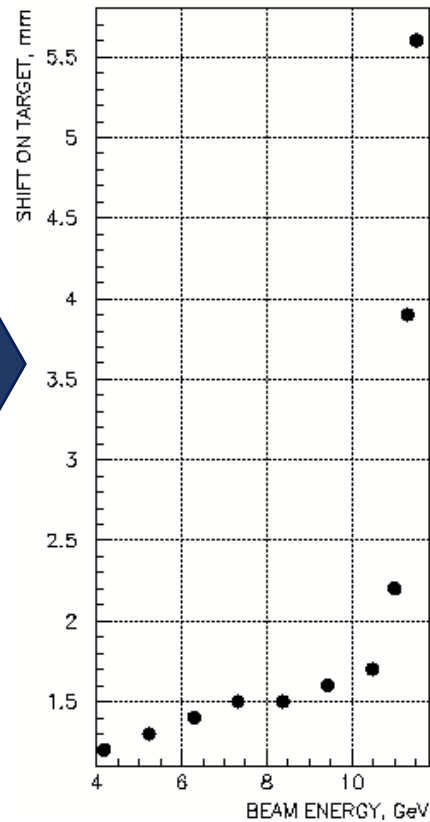
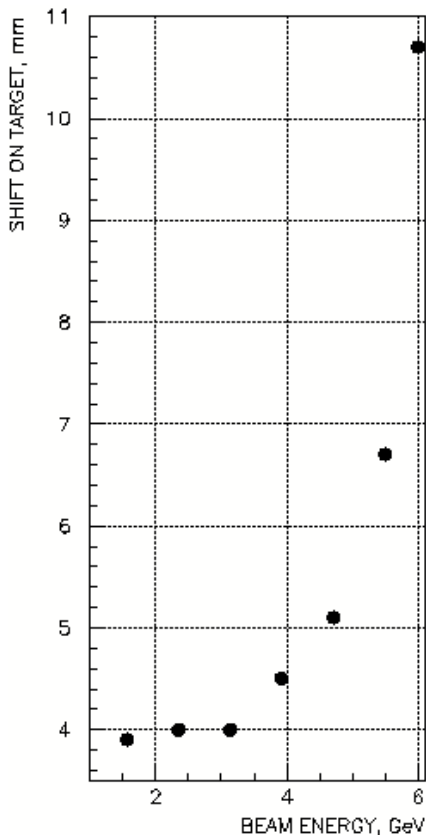


Quadrupole Magnets



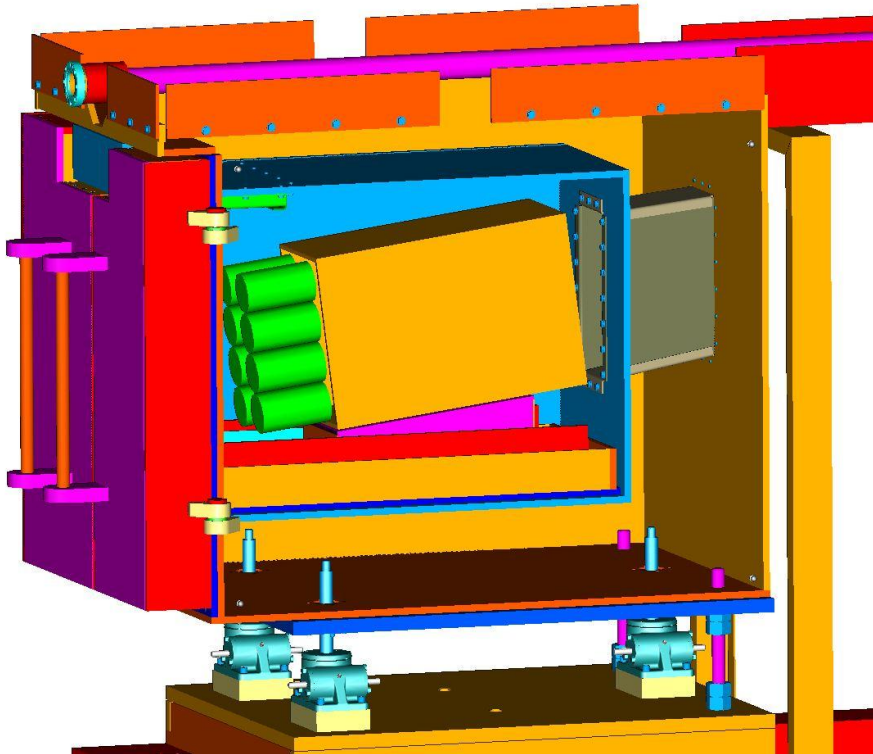
Magnetic properties of Q0 and Q1 Møller quads have been measured (Ken Bagget group)

Møller Dipole Magnet



- Shielding pipe inserted into the dipole to increase the dipole field screening
- To keep the same dipole power supply Møller electrons bending angle reduced from 10^0 to 7.3^0

Møller Detector and Shielding Box



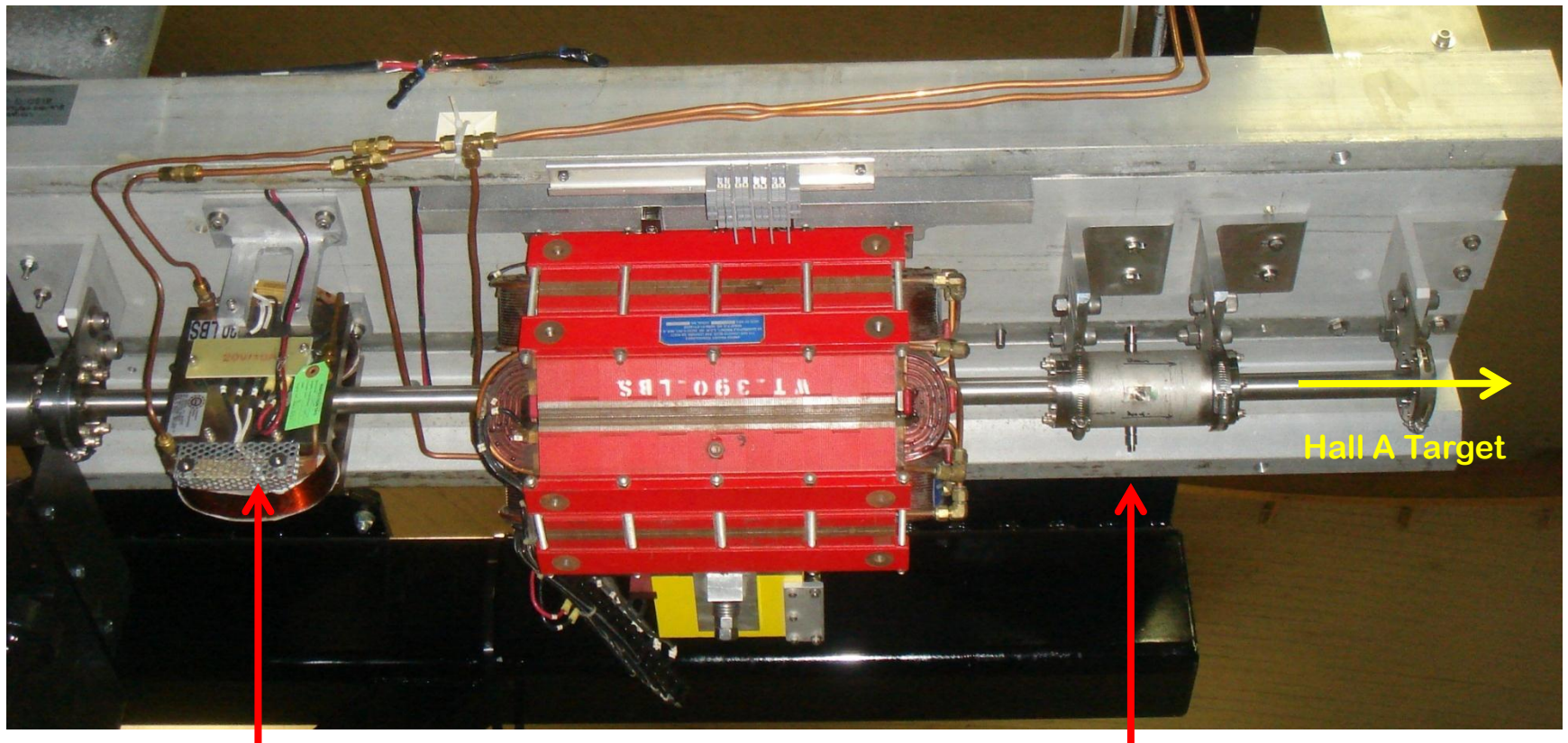
New detector shielding box:

- Lifted up on 7cm for the dipole angle change from 10° to 7.3°
- New doors design
- Improved design of the window to the detector
- Adjusted mounting angle to the detector from 10° to 7.3°
- Top of the box extended to safely contain lead shielding bricks

Upgrade of the aperture detector:

- Add LED
- Better wrapping
- Solid PMT to light guide connection
- New detector holder

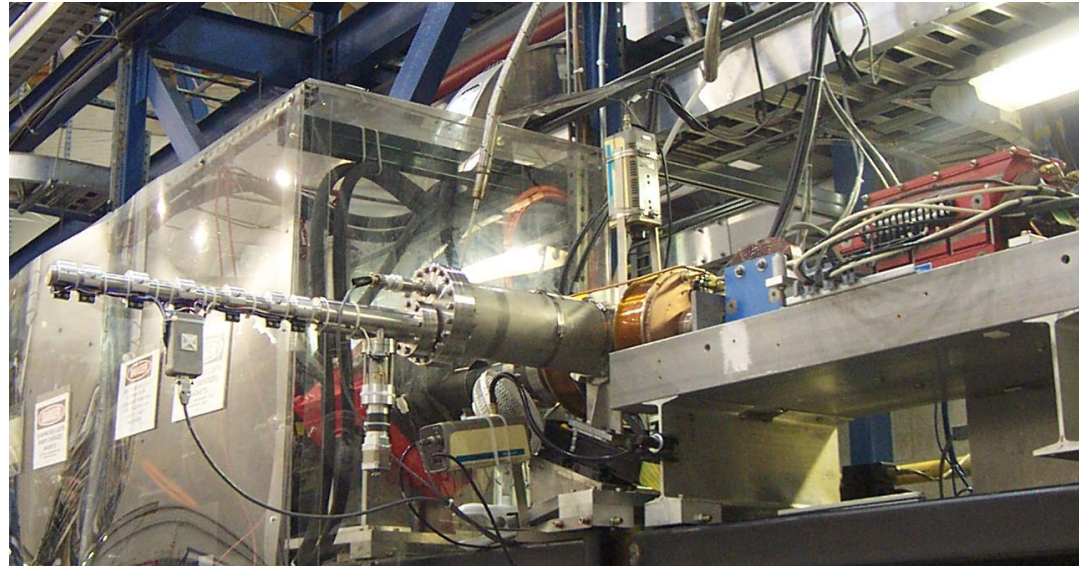
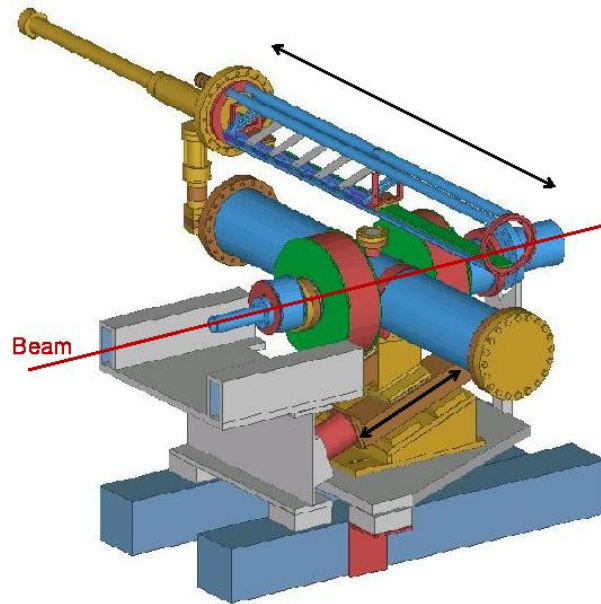
New Girdor Design



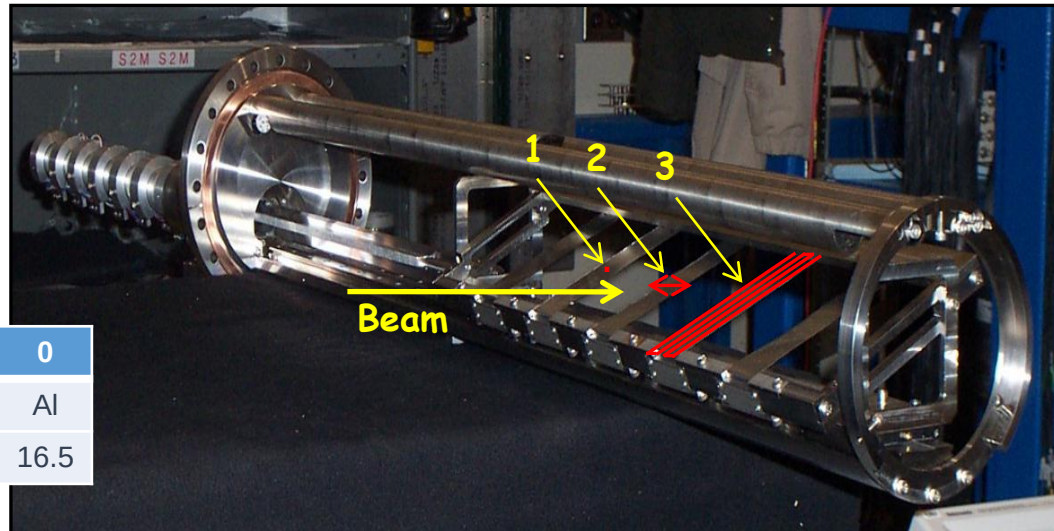
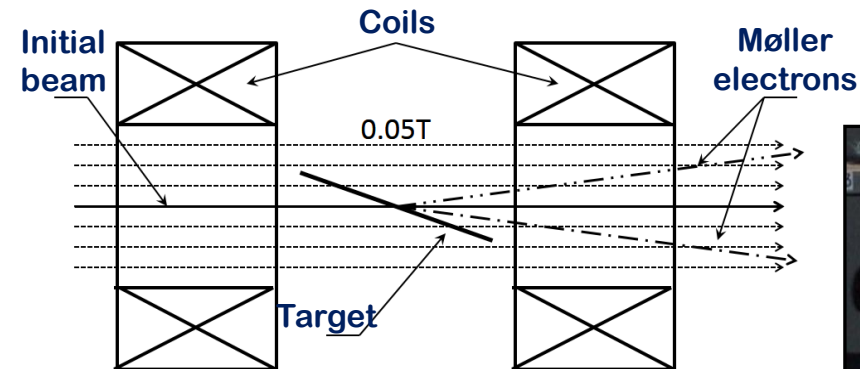
New vertical corrector *MBD1H04*
*To compensate the beam shift after
the Møller dipole*

New BPM *IPM1H04*
*For better beam position
adjustment on the Møller
target*

Low Field Target



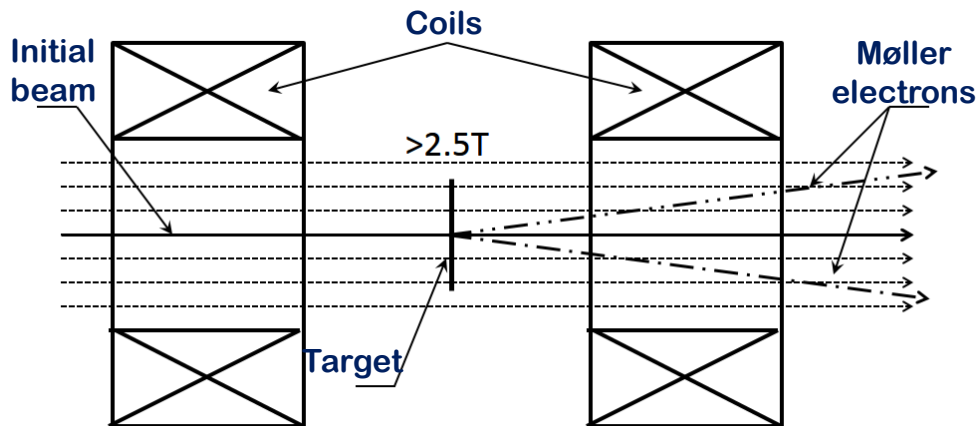
Operating since 2005



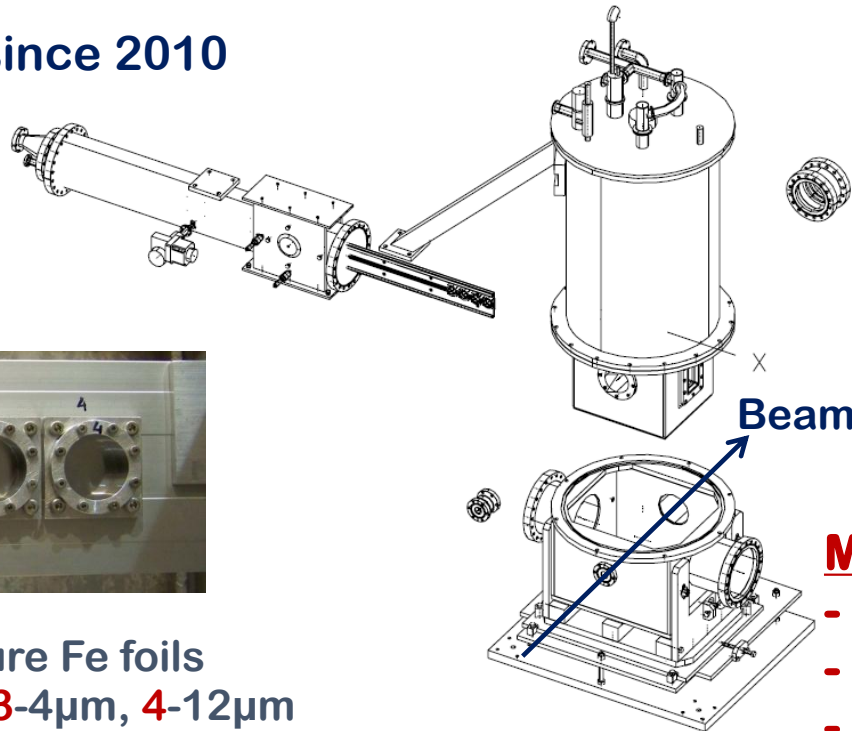
Position	6	5	4	3	2	1	0
Material	park	SM*	Fe	Fe	SM	SM	Al
Thickness, μm	-	6.8	9.3	14.3	29.4	13.0	16.5

*- Supermendure

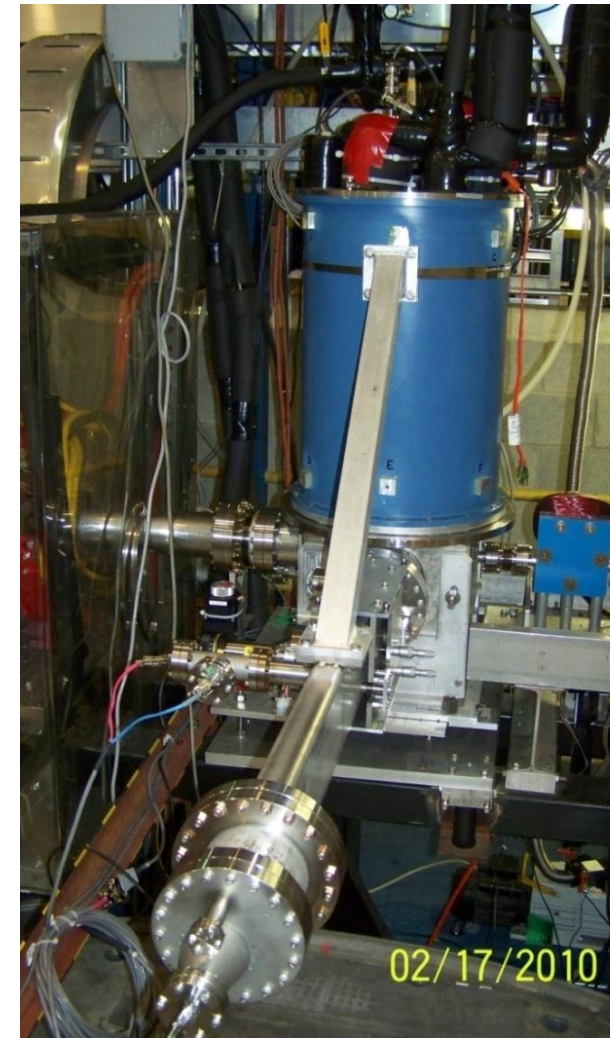
High Field Target



Operating since 2010



Targets: pure Fe foils
1-1 μ m, 2-1 μ m, 3-4 μ m, 4-12 μ m



Must be done:

- New field map
- Field and target alignments
- Feeding cryo line

Møller Polarimeter Systematic Errors

Variable	"Low" field	"High" field 3.9T
<i>Target polarization</i>	1.5%	0.35%
<i>Analyzing power</i>	0.3%	0.3%
<i>Levchuk effect</i>	0.2%	0.3%
<i>Target temperature</i>	0.02%	0.02%
<i>Dead time</i>	0.3%	0.3%
<i>Background</i>	0.3%	0.3%
<i>High beam current</i>	0.2%	0.2%
<i>Others</i>	0.5%	0.5%
<i>Total</i>	1.7%	0.9%

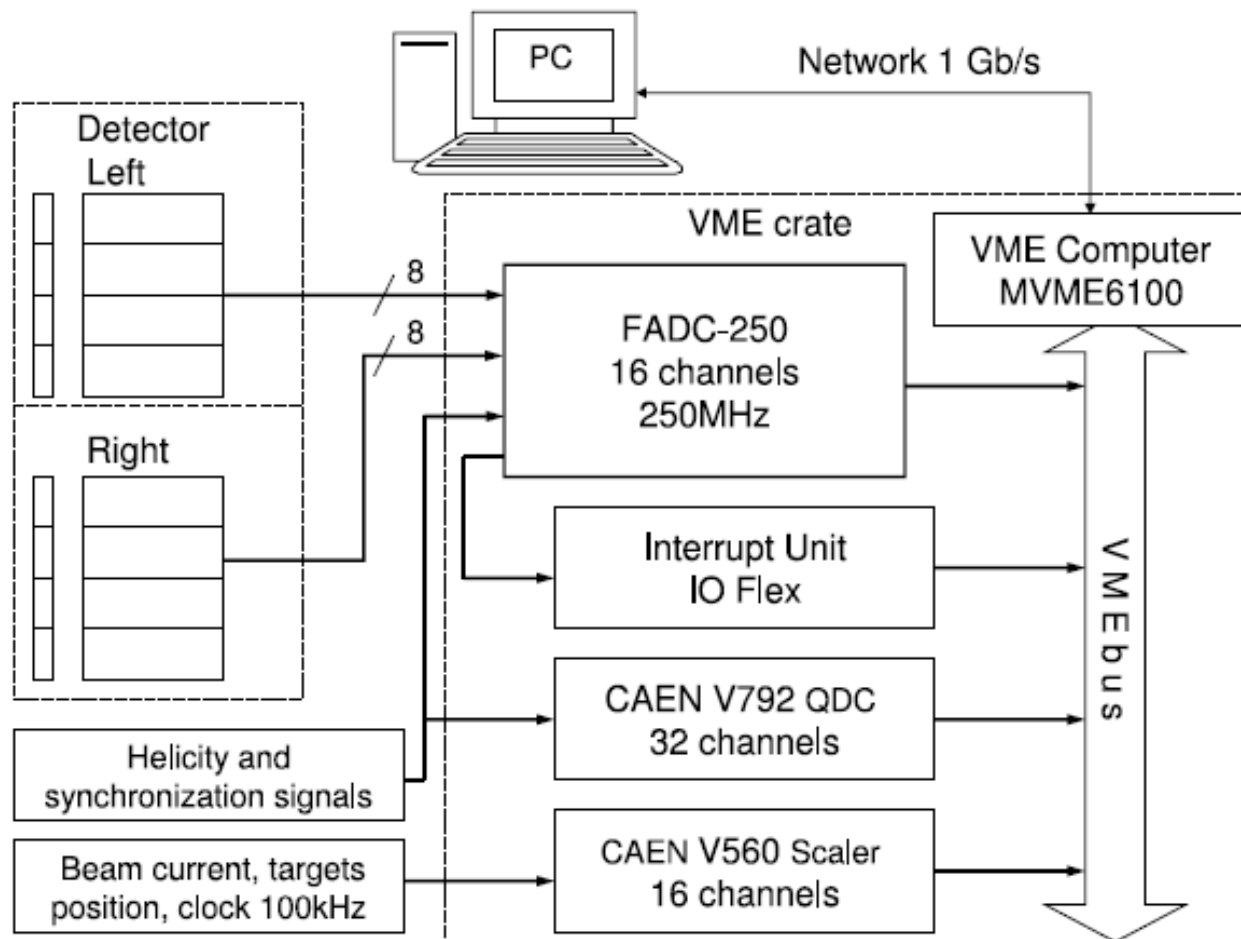
Old DAQ

Operating since 1997.

Upgrade is required:

- **to replace the old PLU module LeCroy-2365 that is not available in stock anymore**
The module CAEN V1495 has the following parameters:
 - Input bandwidth » 200 MHz;
 - 2 input ports x32 bits;
 - 1 output port x32 bits;
 - 2 input/output front LEMO connectors;
 - The FPGA "User" can be reprogrammed by the user using custom logic functions.
- **to increase bandwidth (up to 200 MHz) of the detector system**
 - PLU module LeCroy-2365, bandwidth < 75 MHz, CAMAC replaced with PLU module based on CAEN V1495 board (bandwidth 200 MHz, VME);
 - Discriminator Ortec-TD8000, input rate < 150 MHz, CAMAC replaced with P/S 706 (300 MHz, NIM), modified for remote threshold setup with DAC type of VMIC4140;
- **to reduce readout time from ADC and TDC modules**
 - ADC LeCroy 2249A, 12 channels, CAMAC replaced with QDC CAEN V792 (32 channels, VME);
 - TDC LeCroy 2229, CAMAC replaced with TDC V1190B (64 channels, 0.1 ns, VME).

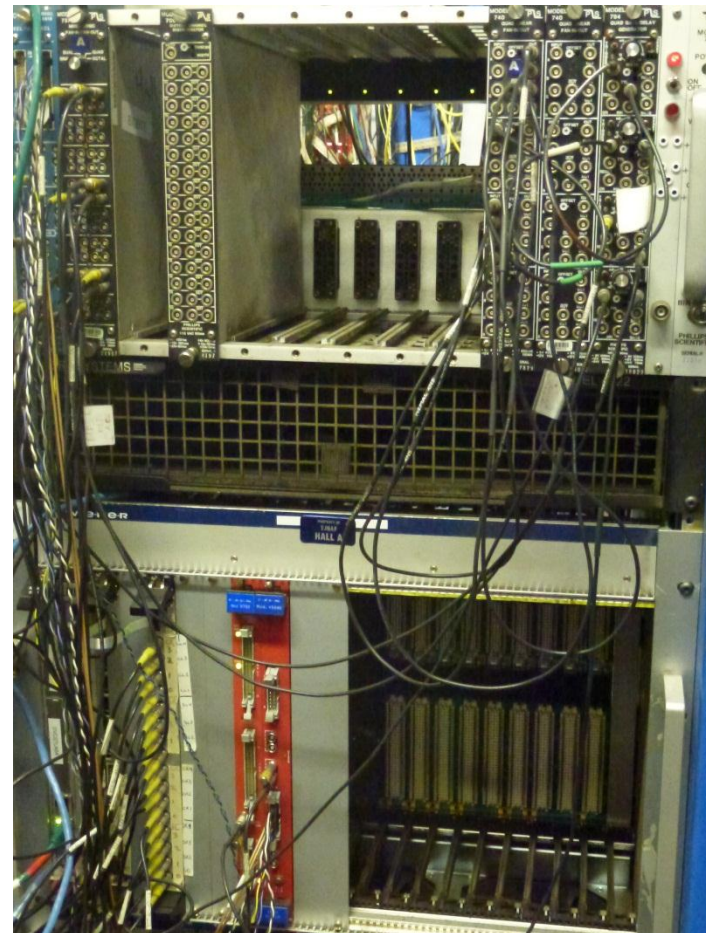
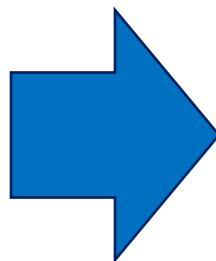
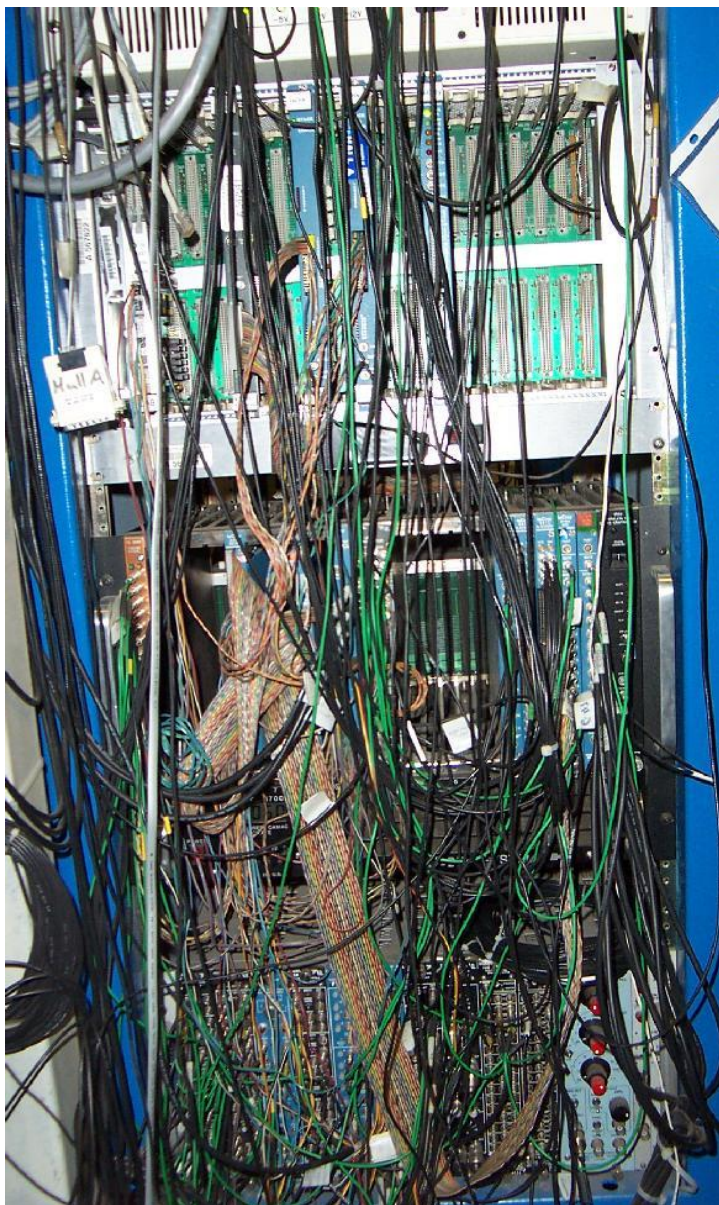
New DAQ based on FADC



The DAQ based on the FADC generates two types of triggers:

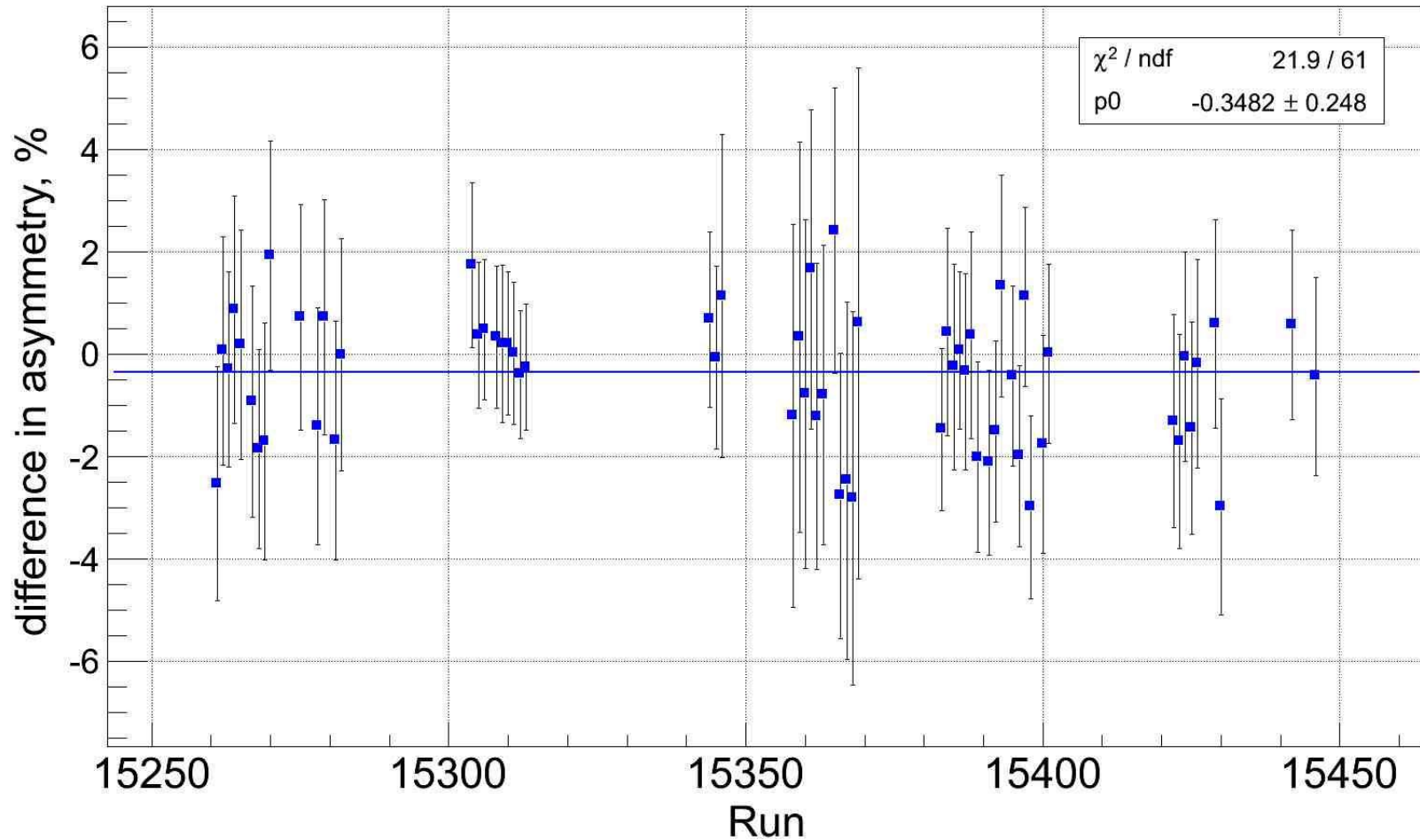
1. Helicity flipping triggers (integral mode / scalers);
2. Data triggers (single events).

Old DAQ vs. FADC DAQ



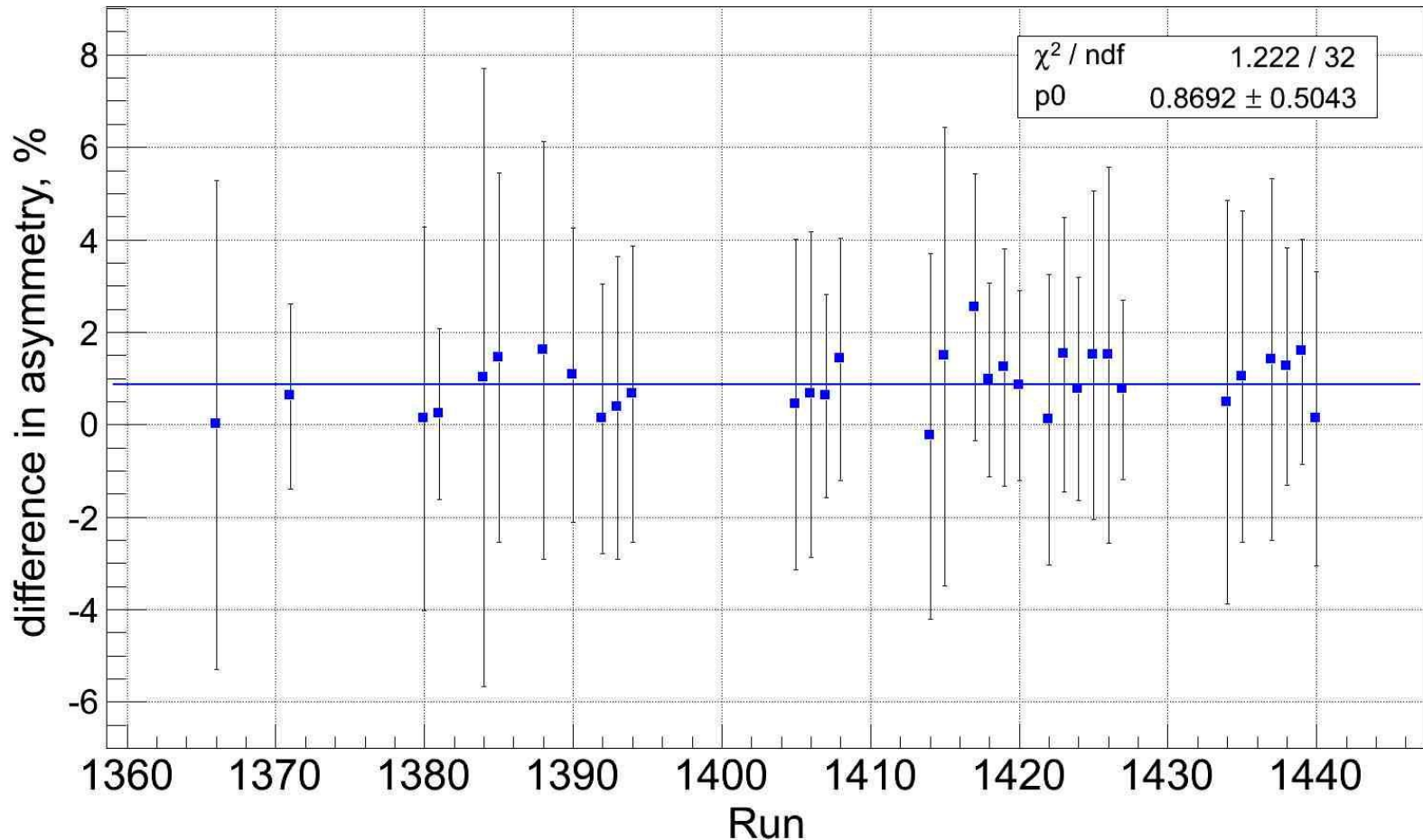
Old DAQ vs. FADC DAQ

Difference between Asymmetry of FADC and Old DAQ



FADC: SCALERS vs. DATA TRIGGERS

Difference between Asymmetry of scalers and data triggers (FADC)



SUMMARY

Summary - I:

Hall A Møller Polarimeter after 11GeV upgrade is ready for commissioning with beam

Summary-II:

Hall A has unique Møller polarimeter:

- **Only one polarimeter in the world has two polarized electron targets based on two different methods of the target polarization**
- **Polarimeter has two DAQs based on different elementary base with different types of trigger and running in parallel**